Name:_____ Lab Group*: Monday *or* Thursday (*delete inappropriate)

Matriculation No.:_____ Dates of Experiment:_____

Fumehood No:_____

CM3191 (AY2026/27 Semester 1)

INORGANIC Experiment 1

Homogenous Catalysis: Tris(triphenylphosphine)ruthenium(II) Dichloride – Synthesis, Characterization and Catalytic Studies

This lab proforma must be handwritten and its softcopy must be submitted together with the appendices (in ONE file) including the 1H-NMR, IR and ESI-MS spectra, of your products obtained, and your own datasheet(s) for all lab sessions, via Canvas-Assignments by 10:00 a.m. on 18-Sep-2026 (Fri.; week 6).*

1. State the balanced reaction equation for the synthesis of $RuCl_2(PPh_3)_3$ in Session I.

2. Calculate the % yield of your $RuCl_2(PPh_3)_3$ obtained in Session I. *[Note: Detailed calculation steps are required.]*

3. Interpret all the signals obtained in the ESI-MS spectra for your $RuCl_2(PPh_3)_3$ in Session I.

4. Tabulate and assign the IR signals obtained for your $RuCl_2(PPh_3)_3$ in Session I.

5. Use no more than **100** words to comment the ^{1}H-NMR signals obtained in Session II and Session III.

[Notes: (i) Do NOT tabulate your experimental results. (ii) Only the first 100 words will be marked if more than 100-word answer is given.]

6. Calculate the % yields of acetophenone obtained in your Session II and Session III. Is there any implication found from such two results? Explain. *[Note: Detailed calculation steps are required.]*

7. Calculate the TOF of the $RuCl_2(PPh_3)_3$ in the catalytic reaction studied. *[Note: Detailed calculation steps are required.]*

8. Use no more than **80** words to comment the value of TOF obtained in Question 7, compared with literature value, and propose ONE possible for such TOF difference.

[Note: Only the first 80 words will be marked if more than 80-word answer is given.]

END OF LAB PROFORMA